



INTERNATIONAL ISLAMIC UNIVERSITY MALAYSIA
COURSE OUTLINE

Kulliyah / Institute	Engineering																								
Department / Centre	Mechatronics Engineering																								
Programme	B. Eng (Mechatronics) (Honours)																								
Name of Course	Machine Vision																								
Course Code	MCTE 4323																								
Name (s) of Academic staff / Instructor(s)	Dr. Amir Akramin Shafie																								
Rationale for the inclusion of the course / module in the programme	Knowledge of the content of this course is the key component in understanding specialised topic of Mechatronics Engineering.																								
Semester and Year Offered	Every Semester																								
Status	Core																								
Level	6																								
Proposed Start Date	Sem 1 – 2015/2016																								
Batch of Student to be Affected	Sem 1 – 2015/2016																								
Total Student Learning Time (SLT)	<table border="1"> <thead> <tr> <th colspan="3">Face to Face</th><th colspan="2">Assessments</th><th rowspan="2">Independent Learning</th><th rowspan="2">Total Student Learning Time</th></tr> <tr> <th>Lecture</th><th>Tutorial</th><th>Practical</th><th>Midterm</th><th>Final</th></tr> </thead> <tbody> <tr> <td>42</td><td></td><td></td><td>2</td><td>3</td><td>73</td><td>120</td></tr> </tbody> </table>						Face to Face			Assessments		Independent Learning	Total Student Learning Time	Lecture	Tutorial	Practical	Midterm	Final	42			2	3	73	120
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42			2	3	73	120																			
Credit Value / Hours	3																								
Pre-requisites (if any)																									
Co-requisites (if any)	None																								

Course Objectives	The objectives of this course are to: 1. Introduce students to the basic concepts and techniques of machine vision. 2. Expose students to essential elements of image acquisition and image pre-processing. 3. Familiarize students with various algorithms for image processing. 4. Illustrate to students various applications of vision system																																																																																										
Learning Outcomes	Upon completion of the course students should be able to: 1. Develop machine vision systems using specific software tools. 2. Apply modern machine vision algorithm for engineering applications. 3. Apply 2D and 3D machine vision techniques for various engineering applications. 4. Select appropriate techniques for the object recognition for various applications.																																																																																										
Transferable Skills:	<i>Skills and how they are developed and assessed:</i> <table><tr><th>Skills</th><th>Development</th><th>Assessment</th></tr><tr><td>Technical</td><td>Lectures</td><td>Written Assessment</td></tr><tr><td>Analytical</td><td>Projects</td><td>Report</td></tr><tr><td></td><td></td><td></td></tr></table>													Skills	Development	Assessment	Technical	Lectures	Written Assessment	Analytical	Projects	Report																																																																					
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Teaching-Learning and assessment strategy	Lectures, Projects report and Quizzes																																																																																										
Course Synopsis	Machine vision concepts. Image acquisition. Lighting. Image formation. Image conversion. Cameras. Vision sensors. Image processing and analysis. Three-dimensional machine vision techniques. Image enhancement. Edge detection. Image interpretation.																																																																																										
Mode of Delivery	Lecture, Tutorial																																																																																										
Assessment Methods and Type/Course Assesment State weightage of each type of assessment.	<table><tr><th>LO</th><th colspan="10">Method</th><th colspan="2">%</th></tr><tr><td>1,2,3,4</td><td colspan="10">Quiz</td><td colspan="2">10</td></tr><tr><td>1,2,3,4,5</td><td colspan="10">Assignments</td><td colspan="2">20</td></tr><tr><td>1,2,3</td><td colspan="10">Mid-term Examination</td><td colspan="2">30</td></tr><tr><td>1,2,3,4</td><td colspan="10">Final Examination</td><td colspan="2">40</td></tr><tr><td></td><td colspan="10">Total</td><td colspan="2">100</td></tr></table>													LO	Method										%		1,2,3,4	Quiz										10		1,2,3,4,5	Assignments										20		1,2,3	Mid-term Examination										30		1,2,3,4	Final Examination										40			Total										100	
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Content outline of the course / module and the SLT per topic																
Weeks	Topics									Learning Hours	Task/Reading					
1	Introduction: Machine Vision. Image Geometry. Sampling and Quantization. Image Definitions. Level of Computation. Road Map.									3	Chapter 1					
2	Image representation: Image digitization, digital image properties, color images, cameras									6	Chapter 2					
3	Image formation Physics: Radiometric measurement, capture and optics, lens aberration and distortion, radiometric image capture, surface reflectance									6	Chapter 3					
4	Data structure for image analysis: Matrices, chains, topological data structures, relational data structures, pyramids, quadtrees									8	Chapter 4					
5,6	Image Pre Processing: Pixel brightness transformation, geometric transformations, local pre-processing, image restoration									8	Chapter 5					
7	Segmentation: Thresholding, edge-based segmentation, region-based segmentation, matching, evaluation issues, Texture									8	Chapter 6					
8,9	Object Recognition: Region identification, contour based shape representation, region based shape representation, shape classes, knowledge representation, graph matching, optimization techniques									8	Chapter 9					
10	Image Understanding: control strategies, RANSAC, point distribution models, scene labeling, semantic image segmentation and understanding									8	Chapter 10					
11,12	3D Vision: 3D vision task, projective geometry, scene reconstruction from multiple views, stereo vision system, trifocal sensor, 3d information from radiometric measurements, Case studies									12	Chapter 11					



13,14	Dynamic Vision: Change Detection. Segmentation Using Motion. Motion Correspondence. Image Flow. Segmentation using Moving Camera. Tracking. Shape from Motion., Case Studies	12	Chapter 16
	Assessment (Mid Term & Final)	8	
	Total SLT	120	

Required references supporting the course

The reference lists shall be presented in accordance with APA bibliographic practices and in alphabetical order.

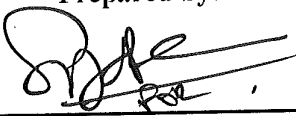
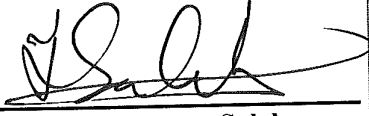
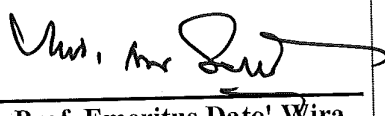
Sonka M. et. al., (2008), *Image Processing, Analysis and Machine Vision*. Thomson Learning.

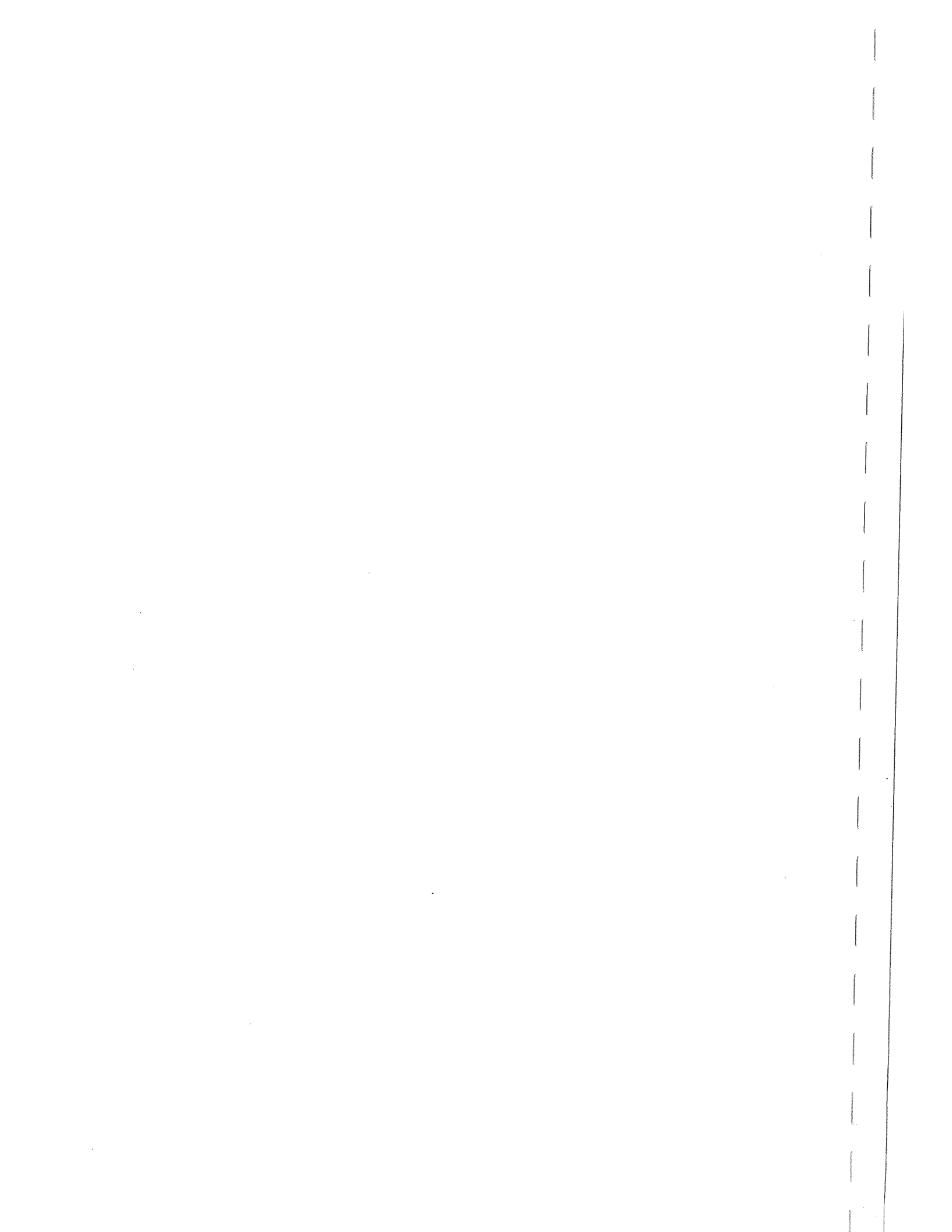
Recommended references supporting the course

Crane, R., (1997). *A simplified Approach to Image Processing (classical & modern techniques in C)*. Prentice Hall .

Petrou, M., & Bosdogianni, P., (2000). *Image Processing. The fundamentals* John Wiley & Sons.

Sonka, M., Hlavac, V., & Boyle, R., (1994). *Image Processing, Analysis and Machine Vision*. Chapman & Hall Computing.

Prepared by:  Dr. Amir Akramin Shafie Assistant Professor Kulliyyah of Engineering	Checked by:  Dr. Tanveer Saleh Assistant Professor Head of Department Mechatronics Engineering	Approved by:  Prof. Emeritus Dato' Wira Ir. Dr. Md Noor bin Salleh Dean Kulliyyah of Engineering
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Kulliyyah Programme Outcomes and the relation between KOE PO with outcomes from EAC, MQF domain, MOHE domain and Soft Skills.

KOE PO	EAC	MQF Domain	MOHE Domain	Soft Skills
1. Engineering Knowledge (T) - Apply knowledge of mathematics, sciences, engineering fundamentals and specialization to solve complex engineering problems.	1	1 & 6	1	-
2. Problem Analysis (T) – Identify, formulate, perform relevant literature review and analyze complex engineering problems , and reaching substantiated conclusions using first principles of mathematics, natural sciences and engineering sciences.	2	1 & 6	1	1
3. Design/Development of Solutions (A) – Design solutions whilst exhibiting innovativeness , for complex engineering problems and design systems, components or processes that meet specified needs; with appropriate consideration of cost, sustainability issues, environmental impact, public health and safety, engineering ethics as well as cultural and social needs.	3	2, 3 & 6	2	1
4. Investigation (D) - Conduct investigation on complex problems whilst displaying creativity , by using research-based knowledge and method, including design of experiments, analysis and interpretation of data , and synthesis of information to provide valid conclusions.	4	2 & 6	2, 3	1
5. Modern Tool Usage (A & D) - Create and apply appropriate techniques, resources and modern engineering/IT tools , which includes making prediction and modelling of the complex engineering activities with understanding of limitations.	5	6 & 7	7	
6. The Engineer and Society (ESSE) - Apply reasoning based on contextual knowledge to assess societal, health, safety, legal, cultural, contemporary issues , and the consequent responsibilities relevant to professional engineering practices.	6	3 & 4	5	4
7. Environment and Sustainability (ESSE) - Understand the impact of professional engineering solutions in societal, global, and environmental contexts and demonstrate knowledge of and need for sustainable development.	7	3 & 4	5	4
8. Ethics (ESSE) –Apply professional ethics with Islamic values and commit to responsibilities and norms of professional engineering code of practices.	8	3 & 4	6	4
9. Communication (S) - Communicate effectively within the engineering community and with the society at large, which include but not limited to writing effective reports and documentation, delivering effective presentation as well as giving and receiving clear instructions.	9	5 & 7	4, 7	2

10. Individual and Team Work (S) - Able to function effectively both as an individual or member of a team, or a leader in a diversified multi-disciplinary team settings.	10	5 & 8	5, 8	3
11. Life Long Learning (S) -Recognize the need for, and have the preparation and ability to engage in independent and life-long learning in the broadest context of technological change.	11	7	7	5
12. Project Management and Finance (S) - Demonstrate and apply engineering management and financial principles into one's work which include being an effective member/leader in projects with multidisciplinary settings and identify opportunities of entrepreneurship.	12	8	5, 8, 9	6, 7

The program learning outcomes (PO) are grouped into 5 general areas to identify the nature of the skills and capability involved. These groups are:

1. Technical (T) – essential capabilities related to traditional scientific and engineering knowledge
2. Analysis (A) – creatively working with available data and engineering tools and fundamental knowledge to correctly solve basic problem
3. Design (D) – being able to perceive the best solution for both small scale and large scale project by involving all required basic problems
4. Ethics, Safety, Society and Environment (ESSE) - giving appropriate consideration to matters pertaining to professionalism and ethics, safety, local and global society and the environment
5. Work skills (S) – being an effective communicator and effective member of a team and to appreciate the need to continuously acquired skills and abilities.

LEARNING OUTCOMES OF OTHER DOMAINS

MQF learning outcomes domains:	MOHE Domain Learning Outcomes	
1. knowledge 2. practical skills 3. social skills and responsibilities 4. values, attitudes and professionalism 5. communication, leadership and team skills, 6. problems solving and scientific skills 7. information management and lifelong learning skills; and 8. managerial and entrepreneurial skills	MOHE learning outcomes domains: 1. Knowledge in Specific Area-Content 2. Practical Skills 3. Critical Thinking and Scientific Skills 4. Communication Skills 5. Social Skills, Teamwork and Responsibilities 6. Values, Ethics, Moral and Professionalism 7. Information Management and Life Long Learning 8. Management and Entrepreneurship 9. Leadership Skills	Soft Skills Learning Outcomes : 1. Critical Thinking and Problem-solving Skills 2. Communication Skills 3. Teamwork Skills 4. Ethics & Moral Professionalism 5. Life-long Learning and Information Management 6. Entrepreneurial Skills 7. Leadership Skills